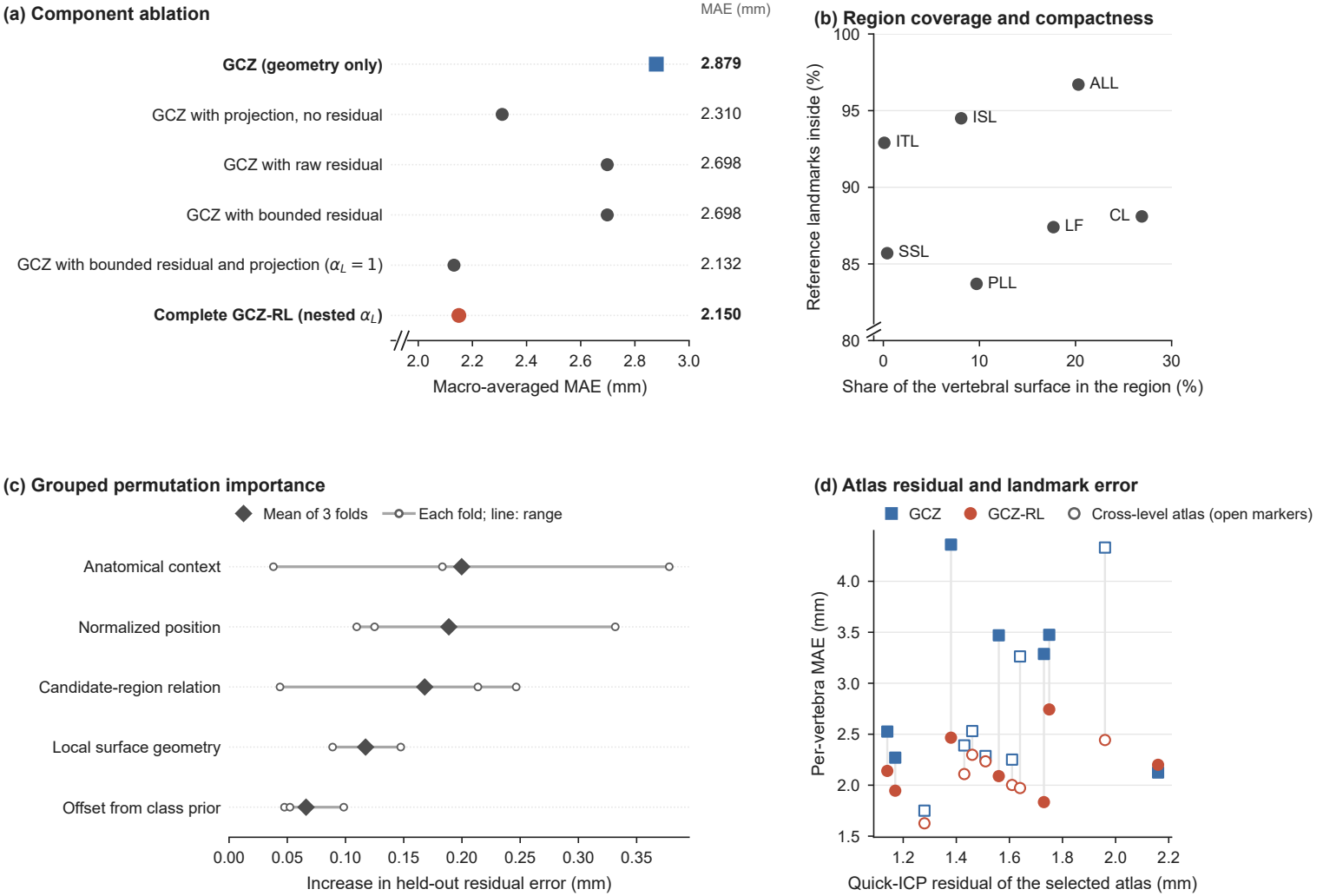

(a) Component ablation
MAE (mm)
GCZ (geometry only)
2.879
GCZ with projection, no residual
2.310
GCZ with raw residual
2.698
GCZ with bounded residual
2.698
GCZ with bounded residual and projection ($\alpha_L = 1$)
2.132
Complete GCZ-RL (nested α_L)
2.150
2.0
2.2
2.4
2.6
2.8
3.0
Macro-averaged MAE (mm)
(b) Region coverage and compactness
Reference landmarks inside (%)
100
95
90
85
80
ALL
ISL
ITL
LF
CL
SSL
PLL
0
10
20
30
Share of the vertebral surface in the region (%)
(c) Grouped permutation importance
Mean of 3 folds
Each fold; line: range
Anatomical context
Normalized position
Candidate-region relation
Local surface geometry
Offset from class prior
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Increase in held-out residual error (mm)
(d) Atlas residual and landmark error
GCZ
GCZ-RL
Cross-level atlas (open markers)
Per-vertebra MAE (mm)
4.0
3.5
3.0
2.5
2.0
1.5
1.2
1.4
1.6
1.8
2.0
2.2
Quick-ICP residual of the selected atlas (mm)